Height Expansion Coefficients of Dominant Rational Maps

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Abstract

Let X be a projective variety over $\overline{\mathbb{Q}}$, and let H be an ample Cartier divisor on X. We study Silverman' s height expansion coefficient $\mu_{X,H}$ for dominant rational self-maps. This coefficient measures the phenomenon of height expansion without functoriality. We study its dynamical behavior using thresholds and give criteria relating these maps to polarized endomorphisms.

For $X = \mathbb{P}^N$, we determine the exact value of the infimum

$$\underline{\mu}_d(\mathbb{P}^N) = d^{-(N-1)} \text{ when } N \geq 2,$$

answering the questions posed by Silverman. Furthermore, by constructing a pair of étale morphisms and using a parameter-space argument involving an incidence correspondence, we give an equivalent characterization of the equality case.

1 Introduction

In this paper, a variety is an integral, separated scheme of finite type over a field k. We assume that all dominant rational self-maps of projective varieties are defined over $\overline{\mathbb{Q}}$ and write $\mathbb{P}^n := \mathbb{P}^n_{\overline{\mathbb{Q}}}$ and $\mathbb{A}^n := \mathbb{A}^n_{\overline{\mathbb{Q}}}$.

For a morphism on projective space, the Weil height (see [4, §B.2]) satisfies an exact functorial growth relation; for dominant rational maps, the corresponding growth phenomenon for heights is more complicated. Silverman has established the following result:

Theorem 1.1 ([17, Theorem 1])**.** *Let $\varphi : W \dashrightarrow V$ be a dominant rational map between quasi-projective varieties over $\overline{\mathbb{Q}}$, and assume that $\dim V = \dim W$. Fix height functions h_V and h_W on V and W, corresponding to ample divisors H_V and H_W. Then there are constants $C_1 > 0$ and C_2, and a nonempty Zariski open set $U \subset W$ such that*

$$h_{V,H_V}(\varphi(P)) \geq C_1 h_{W,H_W}(P) - C_2, \qquad \forall P \in U(\overline{\mathbb{Q}}).$$

The constants C_1 and C_2 depend on V, W, φ, and the choices of h_{V,H_V} and h_{W,H_W}, but are independent of P.

Projective Variety X Case. We begin by working on a projective variety. Following Silverman, we define:

Definition 1.2 (Height Expansion Coefficient)**.** Let X be an N-dimensional projective variety over $\overline{\mathbb{Q}}$, and let H be an ample Cartier divisor on X. Let $\varphi : X \dashrightarrow X$ be a dominant rational self-map. Let $\mathrm{Dom}(\varphi)$ denote its largest open set of definition. We define the *height expansion coefficient* $\mu_{X,H}(\varphi)$ by

$$\mu_{X,H}(\varphi) := \sup_{\substack{\varnothing \neq U \subset \mathrm{Dom}(\varphi) \\ U \text{ is a Zariski open set}}} \liminf_{\substack{P \in U(\overline{\mathbb{Q}}) \\ h_{X,H}(P) \to \infty}} \frac{h_{X,H}(\varphi(P))}{h_{X,H}(P)},$$

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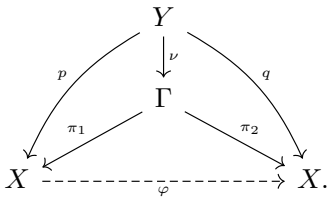
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Equivalently, by the definitions of supremum and infimum, we have:

$$\mu_{X,H}(\varphi) = \sup\left\{ a \geq 0 : \begin{array}{l} \text{there exist a nonempty Zariski open set } U \subset \mathrm{Dom}(\varphi) \\ \text{and a constant } C \geq 0 \text{ such that, for all } P \in U(\overline{\mathbb{Q}}), \\ h_{X,H}(\varphi(P)) \geq a h_{X,H}(P) - C \end{array} \right\}.$$

By virtue of [21, Proposition 3.16(i)] and [13, Proposition 2.3], $h_{X,H}$ is unbounded on any Zariski dense open subset. The height expansion coefficient is not the same as the height growth rate. An example is given in Remark 5.7. Moreover, Theorem 2.1 shows that the height expansion coefficient is invariant under normalization; hence we may assume that X is a normal projective variety.

Normalization Model. Let $\Omega = \mathrm{Dom}(\varphi)$, and set $\Gamma = \overline{\mathrm{Graph}(\varphi|_\Omega)} \subset X \times X$. Let Y be the normalization of Γ, and denote the normalization morphism by $\nu : Y \to \Gamma$. Let $\pi_1, \pi_2 : \Gamma \to X$ be the respective projections, and define $p := \pi_1 \circ \nu$ and $q := \pi_2 \circ \nu$:



Let H be an ample Cartier divisor on X. Then $A := p^*H$ and $B := q^*H$ are nef and big Cartier divisors on Y, and $A + B$ is ample on Y.

For dominant morphisms, Lee [9, Theorem 5.7] gave a pseudo-effective threshold upper bound for height expansion coefficient; Lee also defined the ample threshold [10, Definition 1.1] which differs from the Definition 1.2. For dominant rational self-maps, we use the normalization model above throughout the paper and obtain the corresponding threshold equality:

Proposition 1.3 (Pseudo-effective Threshold)**.** $\mu_{X,H}(\varphi) = \sup\{t \geq 0 : B - tA \in \mathrm{PsEff}(Y)\}$.

Furthermore, given another dominant rational self-map $\psi : X \dashrightarrow X$, Silverman [17, Remark 12] asks about the relations among $\mu_{X,H}(\varphi)$, $\mu_{X,H}(\psi)$, $\mu_{X,H}(\varphi \circ \psi)$ and $\mu_{X,H}(\varphi^n)$. We describe these relations and prove that

Theorem 1.4. *Let $\varphi : X \dashrightarrow X$ be a dominant rational self-map. Then*

$$\lim_{n\to\infty} \mu_{X,H}(\varphi^n)^{1/n} = \frac{\lambda_N(\varphi)}{\lambda_{N-1}(\varphi)},$$

where $\lambda_i(\varphi)$ denotes the i-th dynamical degree of φ. The ratio $\lambda_N(\varphi)/\lambda_{N-1}(\varphi)$ is the smallest *cohomological Lyapunov multiplier* of φ in the terminology of Xie [20]. For related results, see [18], [22], [12]. In addition, we obtain the inequality

$$0 < \mu_{X,H}(\varphi) \leq \frac{\deg(q)(H^N)}{B^{N-1} \cdot A} \leq \deg(q)^{1/N},$$

and investigate how the equality conditions relate to finite or polarized endomorphisms in Theorem 3.3. In a related direction, for a surjective endomorphism f, we establish a criterion for the existence of a polarized iterate in Theorem 3.4. A more refined growth estimate for $\mu_{X,H}$ is given in Corollary 3.5.

Projective Space $\mathbb{P}^N$ Case. In particular, when $X = \mathbb{P}^N$, $\mu_{\mathbb{P}^N,H}$ does not depend on the choice of the ample divisor H, we therefore write $\mu_{\mathbb{P}^N}(\varphi) := \mu_{\mathbb{P}^N,H}(\varphi)$. We define

$$\underline{\mu}_d(\mathbb{P}^N) := \inf_{\substack{\varphi:\mathbb{P}^N \dashrightarrow \mathbb{P}^N \text{ dominant} \\ \deg\varphi = d}} \mu_{\mathbb{P}^N}(\varphi).$$

And we prove the following:

Theorem 1.5 (Uniform Lower Bound). *Every dominant rational map $\varphi : \mathbb{P}^N \dashrightarrow \mathbb{P}^N$ with $\deg\varphi = d$ and $N \geq 2$, satisfies $\mu_{\mathbb{P}^N}(\varphi) \geq d^{-(N-1)}$, hence $\underline{\mu}_d(\mathbb{P}^N) \geq d^{-(N-1)}$.*

When $N = 1$, $h(\varphi(P)) = dh(P) + O(1)$, $\mu_{\mathbb{P}^1}(\varphi) = d$. Combining the upper bound $\underline{\mu}_d(\mathbb{P}^N) \leq d^{-(N-1)}$ [17, Proposition 9], we obtain the following exact value of $\underline{\mu}_d(\mathbb{P}^N)$:

$$\underline{\mu}_d(\mathbb{P}^N) = \begin{cases} d, & N = 1, \\ d^{-(N-1)}, & N \geq 2. \end{cases}$$

Furthermore, by constructing a pair of étale morphisms and using a parameter-space argument involving an incidence correspondence, we give an equivalent characterization of the equality case:

Theorem 1.6 (Characterization of Equality Case). *For the dominant rational map $\varphi : \mathbb{P}^N \dashrightarrow \mathbb{P}^N$ with $\deg\varphi = d$ and $N \geq 2$, the following conditions are equivalent.*

(1) $\mu_{\mathbb{P}^N}(\varphi) = d^{-(N-1)}$;

(2) *φ is birational, and* $\deg(\varphi^{-1}) = d^{N-1}$;

(3) *φ is birational, and* $\dim \mathrm{Ind}(\varphi) \leq 0$.

In particular, when $N = 2$, $\mu_{\mathbb{P}^2}(\varphi) = 1/d$ if and only if φ is birational.

The possible values of the height expansion coefficient are also of interest. Silverman [17, Question 11] asks whether $\mu_{\mathbb{P}^N}(\varphi)$ is always a rational power of $\deg\varphi$. In fact, Lee's example [8, Example 7.7] already yields a negative answer to this question. In Proposition 5.8, we show that every positive rational number can be realized as the height expansion coefficient of a dominant rational self-map of $\mathbb{P}^2$.

2 Normalization Model and Pseudo-effective Threshold

Throughout this section, let X be a projective variety of dimension $N \geq 1$ over $\overline{\mathbb{Q}}$, and let $\varphi\colon X \dashrightarrow X$ be a dominant rational map. By the normalization invariance established in the following theorem, we may henceforth assume that X is a normal projective variety.

Theorem 2.1 (Invariance under normalization). *Let X be a projective variety of dimension $N \geq 1$ over $\overline{\mathbb{Q}}$, let H be an ample Cartier divisor on X, and let $\varphi : X \dashrightarrow X$ be a dominant rational map. Let $\pi : \widetilde{X} \to X$ be the normalization of X, and set $\widetilde{H} := \pi^*H$ and $\widetilde{\varphi} := \pi^{-1} \circ \varphi \circ \pi$. Then $\widetilde{H}$ is ample and*

$$\mu_{\widetilde{X},\widetilde{H}}(\widetilde{\varphi}) = \mu_{X,H}(\varphi).$$

Proof. The normalization morphism π is finite and birational. Thus $\widetilde{X}$ is a normal projective variety, and $\widetilde{H} = \pi^*H$ is ample [7, Proposition 1.2.13].

Set $\Omega := \mathrm{Dom}(\varphi)$, and choose a nonempty open set $V \subset X$ over which π is an isomorphism. Since φ is dominant, the open set $U_0 := V \cap \Omega \cap (\varphi|_\Omega)^{-1}(V)$ is nonempty. Moreover, $\pi : \pi^{-1}(U_0) \to U_0$ is an isomorphism

and $\pi \circ \widetilde{\varphi} = \varphi \circ \pi$ on $\pi^{-1}(U_0)$. By the functoriality of Weil heights [4, Theorem B.3.2], there exists $C_0 \geq 0$ such that

$$\left|h_{\widetilde{X},\widetilde{H}}(\widetilde{P}) - h_{X,H}(\pi(\widetilde{P}))\right| \leq C_0, \qquad \forall \widetilde{P} \in \widetilde{X}(\overline{\mathbb{Q}}).$$

Let $a \geq 0$, and suppose that there exist a nonempty open set $U \subset \Omega$ and a constant $C \geq 0$ such that $h_{X,H}(\varphi(P)) \geq a h_{X,H}(P) - C$ for all $P \in U(\overline{\mathbb{Q}})$. For any $\widetilde{P} \in \pi^{-1}(U \cap U_0)(\overline{\mathbb{Q}})$, we have

$$h_{\widetilde{X},\widetilde{H}}(\widetilde{\varphi}(\widetilde{P})) \geq h_{X,H}(\varphi(\pi(\widetilde{P}))) - C_0 \geq a h_{\widetilde{X},\widetilde{H}}(\widetilde{P}) - C - (a+1)C_0.$$

By Definition 1.2, $\mu_{\widetilde{X},\widetilde{H}}(\widetilde{\varphi}) \geq a$. Taking the supremum over all such a gives $\mu_{\widetilde{X},\widetilde{H}}(\widetilde{\varphi}) \geq \mu_{X,H}(\varphi)$.

By the same argument, we obtain $\mu_{X,H}(\varphi) \geq \mu_{\widetilde{X},\widetilde{H}}(\widetilde{\varphi})$. □

From now on, unless otherwise stated, X is assumed to be a normal projective variety over $\overline{\mathbb{Q}}$ in this section.

Definition 2.2. Let Y be an n-dimensional normal projective variety. A Cartier $\mathbb{R}$-divisor $D = \sum_i a_i D_i$ is called *effective*, denoted by $D \geq 0$, if $a_i \geq 0$ for every prime divisor D_i. Let $\mathsf{N}_i(Y)_{\mathbb{R}}$ denote the real vector space of i-dimensional algebraic cycles on Y modulo numerical equivalence. Let $\mathsf{N}^i(Y)_{\mathbb{R}} := \mathrm{Hom}_{\mathbb{R}}\big(\mathsf{N}_i(Y)_{\mathbb{R}}, \mathbb{R}\big)$, and $\mathsf{N}^1(Y)_{\mathbb{R}}$ is the space of numerical equivalence classes of Cartier $\mathbb{R}$-divisors. We write $\mathrm{Eff}(Y)$, $\mathrm{Amp}(Y)$, $\mathrm{Nef}(Y)$, $\mathrm{Pseff}(Y)$, and $\mathrm{Big}(Y)$ for the cones in $\mathsf{N}^1(Y)_{\mathbb{R}}$ of *effective*, *ample*, *nef*, *pseudo-effective*, and *big* divisor classes, respectively. For convenience, we do not distinguish divisors from their numerical equivalence classes.

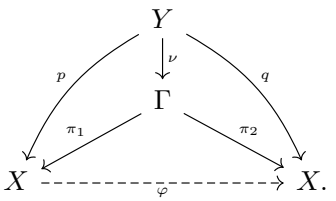
For a Cartier divisor D on Y, define its *volume* by

$$\mathrm{vol}_Y(D) := \limsup_{m \to \infty} \frac{h^0(Y, \mathcal{O}_Y(mD))}{m^n/n!}.$$

Then D is big if and only if $\mathrm{vol}_Y(D) > 0$; if D is nef, then $\mathrm{vol}_Y(D) = (D^n)$ [7, Definition 2.2.31].

Definition 2.3 (Generically Finite Morphism)**.** Let k be a field. If Y, X are varieties over k and $f : Y \to X$ is dominant, then f induces an embedding of function fields $f^* : k(X) \hookrightarrow k(Y)$. If the field extension $[k(Y) : f^*k(X)] < \infty$, then f is called *generically finite*. The morphism f is generically finite if and only if there exists a nonempty affine open set $O \subset X$ such that the restriction morphism $f^{-1}(O) \longrightarrow O$ is finite [19, Lemma 29.52.7].

Definition 2.4 (Normalization Model)**.** Let $\Omega = \mathrm{Dom}(\varphi)$, and set $\Gamma = \overline{\mathrm{Graph}(\varphi|_\Omega)} \subset X \times X$, where the overline denotes the scheme-theoretic closure of the graph. Let Y be the normalization of Γ, and denote the normalization morphism by $\nu : Y \to \Gamma$ [11, Proposition 4.1.22]. Let $\pi_1, \pi_2 : \Gamma \to X$ be the respective projections, and define $p := \pi_1 \circ \nu$ and $q := \pi_2 \circ \nu$:



Let H be an ample Cartier divisor on X. Then $A := p^*H$ and $B := q^*H$ are nef and big Cartier divisors on Y. Since (p, q) is finite and $\mathcal{O}_X(H) \boxtimes \mathcal{O}_X(H)$ is ample, $\mathcal{O}_Y(A + B) \simeq (p, q)^*(\mathcal{O}_X(H) \boxtimes \mathcal{O}_X(H))$ is ample, so $A + B \in \mathrm{Amp}(Y)$.

Lemma 2.5 (Properties of the Graph Model)**.** *Using the notation of Definition 2.4, we have:*

(1) *Y is an N-dimensional normal projective variety.*

(2) *$p, q : Y \to X$ are projective morphisms.*

(3) *p is a birational surjection, and $p : p^{-1}(\Omega) \longrightarrow \Omega$ is an isomorphism. Moreover,*

$$q|_{p^{-1}(\Omega)} = (\varphi|_\Omega) \circ p.$$

(4) *q is a generically finite surjective morphism.*

Proof. By the normalization construction in Definition 2.4 and the commutative diagram, (1) and (2) are clear.

(3) By the definition of Γ, we have $\mathrm{Graph}(\varphi|_\Omega) = \pi_1^{-1}(\Omega) \xrightarrow{\sim} \Omega$ and $\pi_2|_{\pi_1^{-1}(\Omega)} = (\varphi|_\Omega) \circ \pi_1|_{\pi_1^{-1}(\Omega)}$. Since $p = \pi_1 \circ \nu$, $\nu|_{p^{-1}(\Omega)}$ is the normalization of $\pi_1^{-1}(\Omega)$. Thus $p : p^{-1}(\Omega) \to \Omega$ is an isomorphism, and

$$q|_{p^{-1}(\Omega)} = (\varphi|_\Omega) \circ p|_{p^{-1}(\Omega)}.$$

Therefore p is birational. Moreover, since p is projective, its image is closed and contains the dense open set Ω, so p is surjective.

(4) By (3), $q(Y) \supset q(p^{-1}(\Omega)) = \varphi(\Omega)$. Since φ is dominant, $\varphi(\Omega)$ is dense. Since $q(Y)$ is closed and $q(Y) \supset \varphi(\Omega)$, $\overline{\varphi(\Omega)} = X$, the morphism q is surjective, and in particular dominant. Hence it induces an embedding of function fields $q^*k(X) \subset k(Y)$. Moreover,

$$\mathrm{trdeg}_{q^*k(X)}\, k(Y) = \dim Y - \dim X = 0.$$

Thus $k(Y)/q^*k(X)$ is a finitely generated algebraic extension, and hence a finite extension. By Definition 2.3, q is generically finite. □

We first recall the following lemma in a slightly more general form over $\overline{\mathbb{Q}}$. The extension follows readily from the author's proof.

Lemma 2.6 (Lee, [9, Theorem A(5)])**.** *Let W be a projective variety over $\overline{\mathbb{Q}}$, D, E be divisors on W. D is ample. If there exists a dense Zariski open set $V \subset W$ such that*

$$\liminf_{\substack{z \in V(\overline{\mathbb{Q}}) \\ h_{W,D}(z) \to +\infty}} \frac{h_{W,E}(z)}{h_{W,D}(z)} \geq 0,$$

then E is pseudo-effective.

Proof. We can choose a number field $K \subset \overline{\mathbb{Q}}$ and models W_K, D_K, E_K, V_K of W, D, E, V over K. Identifying W with $W_K \times_K \overline{\mathbb{Q}}$, let $\rho : W \to W_K$ be the projection. Then $D = \rho^* D_K$ and $E = \rho^* E_K$, and D_K is ample.

Under the natural identifications $W(\overline{\mathbb{Q}}) = W_K(\overline{\mathbb{Q}})$ and $V(\overline{\mathbb{Q}}) = V_K(\overline{\mathbb{Q}})$, functoriality of heights gives

$$h_{W_K,D_K}(z) = h_{W,D}(z) + O(1), \qquad h_{W_K,E_K}(z) = h_{W,E}(z) + O(1).$$

Moreover, $mD \pm E$ are ample for sufficiently large $m > 0$, so there is a constant $C_0 \geq 0$ such that $|h_{W,E}(z)| \leq m h_{W,D}(z) + C_0$ on $W(\overline{\mathbb{Q}})$. It follows that, as $h_{W,D}(z) \to +\infty$,

$$\frac{h_{W_K,E_K}(z)}{h_{W_K,D_K}(z)} = \frac{h_{W,E}(z)}{h_{W,D}(z)} + O\left(\frac{1}{h_{W,D}(z)}\right).$$

Therefore the hypothesis gives

$$\liminf_{\substack{z\in V_K(\overline{\mathbb{Q}})\\ h_{W_K,D_K}(z)\to+\infty}} \frac{h_{W_K,E_K}(z)}{h_{W_K,D_K}(z)} = \liminf_{\substack{z\in V(\overline{\mathbb{Q}})\\ h_{W,D}(z)\to+\infty}} \frac{h_{W,E}(z)}{h_{W,D}(z)} \geq 0.$$

By [9, Theorem A(5)], $E_K \in \overline{\mathrm{Eff}(W_K)}$. Pullback by ρ^* induces a continuous linear map on numerical spaces and preserves effective divisor classes [2, Example 6.2.9]. Hence

$$E = \rho^* E_K \in \rho^* \overline{\mathrm{Eff}(W_K)} \subseteq \overline{\rho^* \mathrm{Eff}(W_K)} \subseteq \overline{\mathrm{Eff}(W)} = \mathrm{PsEff}(W).$$

□

We now return to the setting of Definition 2.4.

Corollary 2.7. *For every rational number $0 < t < \mu_{X,H}(\varphi)$, we have $B - tA \in \mathrm{PsEff}(Y)$.*

Proof. Write $t = b/a$, where a, b are positive integers. By the equivalent characterization in Definition 1.2, there exist a nonempty open set $U \subset \Omega$ and a constant $C_0 \geq 0$ such that

$$h_{X,H}(\varphi(P)) - t\, h_{X,H}(P) \geq -C_0, \qquad \forall P \in U(\overline{\mathbb{Q}}).$$

For any $y \in p^{-1}(U)(\overline{\mathbb{Q}})$, we have $p(y) \in U \subset \Omega$, so by Lemma 2.5 (3), $q(y) = \varphi(p(y))$. By the properties in [4, Theorem B.3.2],

$$\begin{aligned} h_{Y,aB-bA}(y) &= a h_{X,H}(q(y)) - b h_{X,H}(p(y)) + O(1) \\ &= a\big(h_{X,H}(\varphi(p(y))) - t h_{X,H}(p(y))\big) + O(1) \\ &\geq -aC_0 + O(1). \end{aligned} \tag{2.1}$$

Take any Cartier divisor M on Y with $M \in \mathrm{Amp}(Y)$. As $h_{Y,M}(y) \to +\infty$, the above uniform lower bound gives

$$\liminf_{\substack{y\in p^{-1}(U)\\ h_{Y,M}(y)\to+\infty}} \frac{h_{Y,aB-bA}(y)}{h_{Y,M}(y)} \geq 0.$$

By Lemma 2.6, $aB - bA \in \mathrm{PsEff}(Y)$. Hence $B - tA = \frac{1}{a}(aB - bA) \in \mathrm{PsEff}(Y)$. □

We obtain an equality for dominant rational self-maps, extending Lee's upper bound [9, Theorem 5.7] for dominant morphisms.

Proof of Proposition 1.3. Let $s := \sup\{t \geq 0 : B - tA \in \mathrm{PsEff}(Y)\}$. By the projection formula, $(A^N) = (H^N) > 0$, and $(B^N) = \deg(q)(H^N) > 0$. $A, B \in \mathrm{Nef}(Y) \cap \mathrm{Big}(Y)$. By the openness of the big cone, $B - tA \in \mathrm{Big}(Y)$ for sufficiently small $t > 0$, and hence $s > 0$.

On the other hand, if $B - tA \in \mathrm{PsEff}(Y)$, then since $A \in \mathrm{Nef}(Y)$, $0 \leq (B - tA) \cdot A^{N-1} = (B \cdot A^{N-1}) - t(A^N)$. Therefore,

$$s \leq \frac{(B \cdot A^{N-1})}{(A^N)} < +\infty. \tag{2.2}$$

The closedness of the pseudo-effective cone ensures that $B - sA \in \mathrm{PsEff}(Y)$; $B \in \mathrm{PsEff}(Y)$, so the convexity of the pseudo-effective cone gives

$$\{t \geq 0 : B - tA \in \mathrm{PsEff}(Y)\} = [0, s].$$

Let $0 < t < s$ be any rational number. Since $B - sA \in \mathrm{PsEff}(Y)$ and $A \in \mathrm{Big}(Y)$, $B - tA = B - sA + (s-t)A \in \mathrm{Big}(Y)$. After eliminating denominators, we can choose positive integers a', b' such that $t = a'/b'$ and $H^0\big(Y, \mathcal{O}_Y(b'B - a'A)\big) \neq 0$. Hence, there exists $l \in \mathbb{N}$ and a divisor F_t such that

$$F_t \in \mathrm{Eff}(Y), \qquad l(b'B - a'A) \sim F_t.$$

Let $\Omega = \mathrm{Dom}(\varphi)$, and set a nonempty open set $U_t := \Omega \setminus p(\mathrm{Supp}\, F_t)$. For any $P \in U_t(\overline{\mathbb{Q}})$, since p is surjective, we may choose $y \in Y(\overline{\mathbb{Q}})$ such that $p(y) = P$ and $y \in p^{-1}(\Omega) \setminus \mathrm{Supp}\, F_t$. By Lemma 2.5 (3), we have $q(y) = \varphi(P)$. Thus,

$$\begin{aligned} h_{X,H}(\varphi(P)) - t h_{X,H}(P) &= h_{Y,B}(y) - t h_{Y,A}(y) + O(1) \\ &= \frac{1}{b'} h_{Y,b'B-a'A}(y) + O(1) \\ &= \frac{1}{b'l} h_{Y,F_t}(y) + O(1) \\ &\geq -C_t, \end{aligned}$$

By Definition 1.2, $\mu_{X,H}(\varphi) \geq t$. Letting the rational number $t \uparrow s$, we obtain

$$\mu_{X,H}(\varphi) \geq s. \tag{2.3}$$

On the other hand, let $0 < t < \mu_{X,H}(\varphi)$ be any rational number. By Corollary 2.7, $B - tA \in \mathrm{PsEff}(Y)$, so $t \leq s$. Therefore $\mu_{X,H}(\varphi) \leq s$. Combining the two inequalities, we obtain

$$\mu_{X,H}(\varphi) = s = \sup\{t \geq 0 : B - tA \in \mathrm{Pseff}(Y)\}.$$

□

3 Dynamical Behavior of $\mu_{X,H}(\varphi)$

Throughout this section, let X be a projective variety of dimension $N \geq 1$ over $\overline{\mathbb{Q}}$. Moreover, we may assume that X is a normal projective variety due to Theorem 2.1. Fix an ample Cartier divisor H and a corresponding Weil height $h_{X,H}$. For a dominant rational self-map $\varphi : X \dashrightarrow X$, write $X_\varphi(\overline{\mathbb{Q}})$ for the set of points with well-defined forward orbits under φ.

Retain the notation $Y, p, q, A = p^*H$, and $B = q^*H$ for the normalization model of φ introduced in Definition 2.4. Following [20, §1.1], for any integer $0 \leq i \leq N$, we define the *i-th degree* of φ with respect to H by

$$\deg_{i,H}(\varphi) := \big((p^*H)^{N-i} \cdot (q^*H)^i\big) = (A^{N-i} \cdot B^i).$$

The *i-th dynamical degree* of φ is

$$\lambda_i(\varphi) := \lim_{m\to\infty} \deg_{i,H}(\varphi^m)^{1/m} \geq 1.$$

The limit exists and is independent of the choice of H.

Given another dominant rational self-map $\psi : X \dashrightarrow X$, Silverman [17, Remark 12] asks about the relations among $\mu_{X,H}(\varphi)$, $\mu_{X,H}(\psi)$, $\mu_{X,H}(\varphi \circ \psi)$ and $\mu_{X,H}(\varphi^n)$. We describe these relations and prove that

$$\lim_{n\to\infty} \mu_{X,H}(\varphi^n)^{1/n} = \frac{\lambda_N(\varphi)}{\lambda_{N-1}(\varphi)}.$$

Lemma 3.1. *For any two dominant rational self-maps $\varphi, \psi : X \dashrightarrow X$, we have*

$$\mu_{X,H}(\psi \circ \varphi) \geq \mu_{X,H}(\psi)\mu_{X,H}(\varphi).$$

Proof. Take any $0 < a < \mu_{X,H}(\varphi)$ and $0 < b < \mu_{X,H}(\psi)$. Definition 1.2 gives open sets $U_\varphi \subset \mathrm{Dom}(\varphi)$ and $U_\psi \subset \mathrm{Dom}(\psi)$ such that there exist constants C_φ, C_ψ satisfying

$$h_{X,H}(\varphi(P)) \geq a h_{X,H}(P) - C_\varphi, \qquad \forall P \in U_\varphi(\overline{\mathbb{Q}}),$$

$$h_{X,H}(\psi(Q)) \geq b h_{X,H}(Q) - C_\psi, \qquad \forall Q \in U_\psi(\overline{\mathbb{Q}}).$$

Since φ is dominant, $\varphi^{-1}(U_\psi) = \{P \in \mathrm{Dom}(\varphi) : \varphi(P) \in U_\psi\} \subset \mathrm{Dom}(\varphi)$ is a nonempty open set. Hence $U = U_\varphi \cap \varphi^{-1}(U_\psi) \subset \mathrm{Dom}(\psi \circ \varphi)$ is also a nonempty open set. For any $P \in U(\overline{\mathbb{Q}})$, we obtain

$$h_{X,H}(\psi(\varphi(P))) \geq b h_{X,H}(\varphi(P)) - C_\psi \geq ab h_{X,H}(P) - bC_\varphi - C_\psi.$$

Dividing both sides by $h_{X,H}(P)$ and letting $h_{X,H}(P) \to \infty$, we obtain $\mu_{X,H}(\psi \circ \varphi) \geq ab$ by Definition 1.2. Thus

$$\mu_{X,H}(\psi \circ \varphi) \geq \sup_{\substack{0<a<\mu_{X,H}(\varphi) \\ 0<b<\mu_{X,H}(\psi)}} ab = \mu_{X,H}(\psi)\mu_{X,H}(\varphi).$$

□

Lemma 3.2 (Comparison of the height expansion coefficient and multidegrees)**.** *Let $N \geq 1$ and let $\varphi : X \dashrightarrow X$ be a dominant rational map. Then*

$$\frac{\deg_{N,H}(\varphi)}{N \deg_{N-1,H}(\varphi)} \leq \mu_{X,H}(\varphi) \leq \frac{\deg_{N,H}(\varphi)}{\deg_{N-1,H}(\varphi)}. \tag{3.1}$$

Proof. We retain the notation Y, p, q, A, B from Definition 2.4 corresponding to φ. Then we have

$$(B^N) = \deg_{N,H}(\varphi) > 0, \qquad (A \cdot B^{N-1}) = \deg_{N-1,H}(\varphi) > 0.$$

If $t \geq 0$ and $B - tA \in \mathrm{PsEff}(Y)$, then, since $B \in \mathrm{Nef}(Y)$, arguing as in equation (2.2), we obtain

$$\mu_{X,H}(\varphi) \leq \frac{(B^N)}{(A \cdot B^{N-1})}. \tag{3.2}$$

For any rational number $0 < t < \frac{(B^N)}{N(A \cdot B^{N-1})}$, we have $(B^N) > N(B^{N-1} \cdot tA)$. By Siu's theorem [7, Theorem 2.2.15], $B - tA \in \mathrm{Big}(Y) \subset \mathrm{PsEff}(Y)$. Letting $t \uparrow \frac{(B^N)}{N(A \cdot B^{N-1})}$, we obtain

$$\mu_{X,H}(\varphi) \geq \frac{(B^N)}{N(A \cdot B^{N-1})}.$$

□

Proof of Theorem 1.4. Taking the nth roots in (3.1), we obtain

$$\frac{1}{N^{1/n}} \frac{\deg_{N,H}(\varphi^n)^{1/n}}{\deg_{N-1,H}(\varphi^n)^{1/n}} \leq \mu_{X,H}(\varphi^n)^{1/n} \leq \frac{\deg_{N,H}(\varphi^n)^{1/n}}{\deg_{N-1,H}(\varphi^n)^{1/n}}.$$

Both the left-hand and right-hand sides converge to $\lambda_N(\varphi)/\lambda_{N-1}(\varphi)$. The conclusion follows naturally. □

Theorem 3.3 (Upper bound and equality case)**.** *With the notation of Definition 2.4, set* $m := (B^{N-1} \cdot A)$*. Let* X *be a normal projective variety. Then:*

(1)
$$0 < \mu_{X,H}(\varphi) \leq \frac{\deg(q)(H^N)}{m} \leq \deg(q)^{1/N}.$$

(2) *The equality* $\mu_{X,H}(\varphi) = \deg(q)^{1/N}$ *holds if and only if* φ *extends to a finite morphism* $\widetilde{\varphi}$ *satisfying* $\widetilde{\varphi}^* H \equiv \deg(q)^{1/N} H$*. In this case,* $\deg(q)^{1/N} \in \mathbb{Z}_{>0}$*.*

(3) *If* $\deg(q) > 1$*, the equality* $\mu_{X,L}(\varphi) = \deg(q)^{1/N}$ *holds for some ample Cartier divisor* L *if and only if* φ *extends to a polarized endomorphism.*

Proof. (1) By Definition 2.4, A and B are nef and big. The projection formula gives $(A^N) = (H^N) > 0$ and $(B^N) = \deg(q)(H^N)$. The Khovanskii–Teissier inequality [7, Corollary 1.6.3] and Lemma 3.2 therefore give

$$m^N \geq (B^N)^{N-1}(A^N) = \deg(q)^{N-1}(H^N)^N > 0.$$

and Equation (3.2) gives

$$0 < \mu_{X,H}(\varphi) \leq \frac{(B^N)}{(B^{N-1} \cdot A)} \leq \frac{(B^N)}{(B^N)^{1-1/N}(A^N)^{1/N}} = \left(\frac{(B^N)}{(A^N)}\right)^{1/N} = \deg(q)^{1/N}.$$

(2) $\Longrightarrow$: Suppose that $\mu_{X,H}(\varphi) = \deg(q)^{1/N}$. Then $(B^{N-1} \cdot A) = (B^N)^{1-1/N}(A^N)^{1/N}$. By [1, Theorem D], A and B are numerically proportional; since $(B^N) = \deg(q)(A^N)$, we obtain $B \equiv \deg(q)^{1/N} A$.

As $A + B \in \mathrm{Amp}(Y)$, both A and B are ample. For every integral curve $C \subset Y$, ampleness and the projection formula give $0 < (A \cdot C) = (H \cdot p_* C)$ and $0 < (B \cdot C) = (H \cdot q_* C)$. Thus neither p nor q contracts a curve. Since every positive-dimensional projective fibre contains a curve, p and q are quasi-finite. Properness then implies finiteness of p, q. Since p is birational and X is normal, p is an isomorphism. Hence $\widetilde{\varphi} := q \circ p^{-1}$ is finite and $\widetilde{\varphi}^* H = (p^{-1})^* B \equiv \deg(q)^{1/N}(p^{-1})^* A = \deg(q)^{1/N} H$.

$\Longleftarrow$: Conversely, if φ extends to a finite morphism $\widetilde{\varphi}$ satisfying $\widetilde{\varphi}^* H \equiv \deg(q)^{1/N} H$, then $q = \widetilde{\varphi} \circ p$ and $B - \deg(q)^{1/N} A \equiv 0$. Proposition 1.3 and (1) yield $\mu_{X,H}(\varphi) = \deg(q)^{1/N}$.

In this case, $\deg(q)^{1/N} = (\varphi^* H \cdot H^{N-1})/(H^N) \in \mathbb{Q}$. Its N-th power is the positive integer $\deg(q)$, so $\deg(q)^{1/N} \in \mathbb{Z}_{>0}$.

(3) $\Longrightarrow$: If $\mu_{X,L}(\varphi) = \deg(q)^{1/N}$ for some ample Cartier divisor L, then applying (2) with $H = L$ shows that φ extends to a finite surjective morphism $\widetilde{\varphi}$ satisfying $\widetilde{\varphi}^* L \equiv \deg(q)^{1/N} L$ with $\deg(q)^{1/N} \in \mathbb{Z}_{>1}$. This extension is polarized by [14, Lemma 3.4].

$\Longleftarrow$: Suppose that φ extends to a polarized endomorphism $\widetilde{\varphi}$. Choose an ample Cartier divisor L and an integer $a > 1$ such that $\widetilde{\varphi}^* L \sim aL$. Since $q = \widetilde{\varphi} \circ p$ and p is birational, $\deg(\widetilde{\varphi}) = \deg(\varphi) = \deg(q)$. The projection formula gives

$$a^N (L^N) = ((\varphi^* L)^N) = \deg(\varphi)(L^N) = \deg(q)(L^N).$$

As $(L^N) > 0$, we obtain $a = \deg(q)^{1/N}$. Applied (2) with $H = L$, therefore yields $\mu_{X,L}(\varphi) = \deg(q)^{1/N}$. □

In what follows, we consider the case where X is a normal projective variety and $f : X \to X$ is a surjective endomorphism over $\overline{\mathbb{Q}}$, and let H be still an ample Cartier divisor on X. Inspired by [14, Lemma 3.5], we deduce the following criterion:

Theorem 3.4 (Criterion for a polarized iterate)**.** *Let X be an N-dimensional normal projective variety over $\overline{\mathbb{Q}}$, let $f : X \to X$ be a surjective endomorphism with $\deg f > 1$, and let H be an ample Cartier divisor on X. Then*

$$\inf_{n\geq 1} \frac{\mu_{X,H}(f^n)}{(\deg f)^{n/N}} > 0 \quad \text{if and only if} \quad f^N \text{ is polarized.}$$

In this case, there is an ample Cartier divisor L such that $(f^N)^*L \sim (\deg f)L$.

Proof. If $N = 1$, numerical equivalence is determined by degree, so $(f^n)^*H \equiv \deg(f^n)H = (\deg f)^n H$. Proposition 1.3 therefore gives $\mu_{X,H}(f^n) = (\deg f)^n$ for all $n \geq 1$.

$\Longrightarrow$: Taking $n = 1$ gives $\mu_{X,H}(f) = \deg f > 1$. Theorem 3.3 (3) with $L = H$ then shows that $f = f^N$ is polarized.

$\Longleftarrow$: The formula for $\mu_{X,H}(f^n)$ gives

$$\inf_{n\geq 1} \frac{\mu_{X,H}(f^n)}{(\deg f)^n} = 1 > 0.$$

If $N \geq 2$. $\Longrightarrow$: Choose $0 < c \leq 1$ such that $\mu_{X,H}(f^n) \geq c(\deg f)^{n/N}$ for all $n \geq 1$, and put $D_n := (\deg f)^{-n/N}(f^n)^*H \in \mathrm{Nef}(X) \subset \mathsf{N}^1(X)_{\mathbb{R}}$ for $n \geq 0$.

By Proposition 1.3, $(f^n)^*H - tH$ is pseudo-effective for $0 \leq t \leq \mu_{X,H}(f^n)$. Taking $t = c(\deg f)^{n/N}$ and dividing by $(\deg f)^{n/N}$ yields $D_n - cH \in \mathrm{Pseff}(X)$ for every $n \geq 1$. Intersecting $D_n - cH$ with the nef products $D_n^{N-1-j} \cdot H^j$ for $0 \leq j \leq N-2$ and using the projection formula, we obtain

$$(H^N) = (D_n^N) \geq c(D_n^{N-1} \cdot H) \geq \cdots \geq c^{N-1}(D_n \cdot H^{N-1}).$$

Fix a norm $\|\cdot\|$ on $\mathsf{N}^1(X)_{\mathbb{R}}$. By [14, Lemma 2.12], the functional $E \mapsto (E \cdot H^{N-1})$ is strictly positive on $\mathrm{Nef}(X) \setminus \{0\}$, so it has a positive minimum on the compact set $\{E \in \mathrm{Nef}(X) : \|E\| = 1\}$. Thus $\{D_n : n \geq 0\}$ is bounded and has compact closure in $\mathsf{N}^1(X)_{\mathbb{R}}$.

Denote $E_k := k^{-1}\sum_{j=0}^{k-1} D_j$ and $D_\infty := \lim_{\ell\to\infty} E_{k_\ell}$ for some convergent subsequence E_{k_ℓ} of E_k. The Khovanskii–Teissier inequalities give

$$(E_{k_\ell}^N)^{1/N} = \frac{1}{k_\ell}\sum_{j=0}^{k_\ell-1} \frac{(E_{k_\ell}^{N-1} \cdot D_j)}{(E_{k_\ell}^N)^{(N-1)/N}} \geq \frac{1}{k_\ell}\sum_{j=0}^{k_\ell-1} (D_j^N)^{1/N} = (H^N)^{1/N}.$$

Thus D_∞ is nef and big. Moreover, $f^*D_j = (\deg f)^{1/N} D_{j+1}$, and hence

$$f^*E_{k_\ell} - (\deg f)^{1/N} E_{k_\ell} = \frac{(\deg f)^{1/N}}{k_\ell}\sum_{j=0}^{k_\ell-1}(D_{j+1} - D_j) = \frac{(\deg f)^{1/N}}{k_\ell}(D_{k_\ell} - H) \longrightarrow 0,$$

where the limit follows because the D_{k_ℓ} lie in a compact set. Thus $f^*D_\infty \equiv (\deg f)^{1/N} D_\infty$, and consequently $(f^N)^*D_\infty \equiv (\deg f)D_\infty$. By [14, Proposition 3.6], f^N is polarized.

$\Longleftarrow$: Suppose that f^N is polarized. Choose an ample Cartier divisor L and an integer $m > 1$ such that $(f^N)^*L \sim mL$. By the projection formula,

$$m^N(L^N) = \big(((f^N)^*L)^N\big) = \deg(f^N)(L^N) = (\deg f)^N(L^N).$$

Since $(L^N) > 0$, we have $m = \deg f$ and hence $(f^N)^*L \equiv (\deg f)L$. Choose $a, b > 0$ such that $H - aL$ and $L - bH$ are ample. For every $i \geq 0$,

$$(f^{Ni})^*H - ab(\deg f)^i H \equiv (f^{Ni})^*(H - aL) + a(\deg f)^i(L - bH) \in \mathrm{Amp}(X) \subset \mathrm{Pseff}(X).$$

Proposition 1.3 gives $\mu_{X,H}(f^{Ni}) \geq ab(\deg f)^i$. Writing $n = Ni + r$ with $0 \leq r < N$, supermultiplicativity in Lemma 3.1 gives

$$\frac{\mu_{X,H}(f^n)}{(\deg f)^{n/N}} \geq \frac{\mu_{X,H}(f^{Ni})\mu_{X,H}(f^r)}{(\deg f)^{i+r/N}} \geq ab\,\frac{\mu_{X,H}(f^r)}{(\deg f)^{r/N}}.$$

The assertion follows by taking the infimum, since $\mu_{X,H}(f^r) > 0$ for $0 \leq r < N$. □

Corollary 3.5 (Polynomial refinement of height expansion)**.** *Let X be an N-dimensional normal projective variety over $\overline{\mathbb{Q}}$, and let $f : X \to X$ be a surjective endomorphism. Denote $\eta_{N-1}+1$ be the largest size of the Jordan blocks of $f^*|_{\mathsf{N}^{N-1}(X)_{\mathbb{C}}}$ associated with eigenvalues of modulus $\lambda_{N-1}(f)$. Then*

$$\mu_{X,H}(f^n) \asymp \left(\frac{\lambda_N(f)}{\lambda_{N-1}(f)}\right)^n n^{-\eta_{N-1}}.$$

Proof. By [6, Proposition 5.1], $\deg_{N-1,H}(f^n) \asymp \lambda_{N-1}(f)^n n^{\eta_{N-1}}$. The projection formula gives

$$\deg_{N,H}(f^n) = (H^N)(\deg f)^n = (H^N)\lambda_N(f)^n.$$

Thus Lemma 3.2 yields

$$\mu_{X,H}(f^n) \asymp \frac{\deg_{N,H}(f^n)}{\deg_{N-1,H}(f^n)} = \frac{(H^N)\lambda_N(f)^n}{\deg_{N-1,H}(f^n)} \asymp \frac{(H^N)\lambda_N(f)^n}{\lambda_{N-1}(f)^n n^{\eta_{N-1}}} \asymp \left(\frac{\lambda_N(f)}{\lambda_{N-1}(f)}\right)^n n^{-\eta_{N-1}}.$$

□

4 Exact Value of $\underline{\mu}_d(\mathbb{P}^N)$

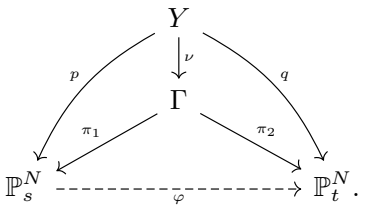
Throughout this chapter, we denote $X = \mathbb{P}^N$ over $\overline{\mathbb{Q}}$, with $N \geq 2$. For a dominant rational self-map $\varphi\colon \mathbb{P}^N \dashrightarrow \mathbb{P}^N$, denote the corresponding copies of $\mathbb{P}^N$ by $\mathbb{P}^N_s$ and $\mathbb{P}^N_t$, and their homogeneous coordinates by $[s_0 : \cdots : s_N]$ and $[t_0 : \cdots : t_N]$, respectively. Suppose

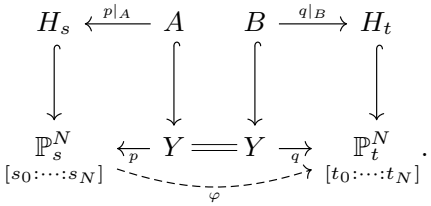
$$\varphi\colon \mathbb{P}^N_s \dashrightarrow \mathbb{P}^N_t, \qquad [s_0 : \cdots : s_N] \longmapsto [F_0(s) : \cdots : F_N(s)]_t,$$

where $F_i \in \overline{\mathbb{Q}}[s_0, \ldots, s_N]$ are homogeneous polynomials of the same degree d, and $\gcd(F_0, \ldots, F_N) = 1$. Then

$$\Omega = \mathrm{Dom}(\varphi) = \mathbb{P}^N_s \setminus V(F_0, \ldots, F_N), \qquad \mathrm{Ind}(\varphi) = V(F_0, \ldots, F_N), \qquad \mathrm{codim}_{\mathbb{P}^N_s} \mathrm{Ind}(\varphi) \geq 2.$$

Definition 4.1. Take the hyperplanes $H_s := V(s_0) \subset \mathbb{P}^N_s$ and $H_t := V(t_0) \subset \mathbb{P}^N_t$. Define $A := p^*H_s \subset Y$ and $B := q^*H_t \subset Y$. The corresponding line bundles satisfy $\mathcal{O}_Y(A) \simeq p^*\mathcal{O}_{\mathbb{P}^N_s}(1)$ and $\mathcal{O}_Y(B) \simeq q^*\mathcal{O}_{\mathbb{P}^N_t}(1)$.





A and B are both nef and big. Denote $m := (AB^{N-1})$. By the projection formula, $(A^N) = 1$, $(A^{N-1}B) = d$. By the Khovanskii–Teissier intersection inequality,

$$d^{N-1} = (A^{N-1}B)^{N-1} \geq (A^N)^{N-2}(AB^{N-1}) = m. \tag{4.1}$$

Definition 4.2 (Hyperplane Parameter Space)**.** The *hyperplane parameter space* $\mathcal{H} := (\mathbb{P}_s^N)^\vee$ is the projective space with homogeneous coordinates $[a_0 : \cdots : a_N]$, where

$$[a_0 : \cdots : a_N] \longleftrightarrow [\ell], \qquad \ell = a_0 s_0 + \cdots + a_N s_N, \quad (a_0, \ldots, a_N) \in \overline{\mathbb{Q}}^{N+1} \setminus \{0\}.$$

We denote the hyperplane corresponding to the equivalence class $[\ell] \in \mathcal{H}(\overline{\mathbb{Q}})$ by $H_{[\ell]} := V(\ell) \subset \mathbb{P}_s^N$.

Definition 4.3. By Definition 2.3 and Lemma 2.5 (4), choose and fix a nonempty Zariski open set $O' \subset \mathbb{P}_t^N$ such that $q^{-1}(O') \to O'$ is finite. Let

$$Z := Y \setminus q^{-1}(O').$$

By Noetherianity, denote the irreducible components of Z of dimension $N-1$ by $R_1, \ldots, R_a$, and fix a closed point $x_i \in R_i$ for each i. Define the open set

$$\mathcal{H}_1 := \mathcal{H} \setminus \bigcup_{i=1}^{a} \big\{[\ell] \in \mathcal{H} : p(x_i) \in H_{[\ell]}\big\}.$$

Lemma 4.4 (Choice of a Hyperplane)**.** *For any $[\ell] \in \mathcal{H}_1$, the divisor $D := p^* H_{[\ell]} \in \operatorname{Eff}(Y)$ is a nonzero Cartier divisor satisfying $D \geq 0$, $D \sim A = p^* H_s$, and every prime divisor component T of D satisfies $\dim q(T) = N-1$.*

Proof. Let $[\ell] \in \mathcal{H}_1$. Then $D := p^* H_{[\ell]}$ is a Cartier divisor satisfying $D \geq 0$ and $D \in \operatorname{Eff}(Y)$, and $\operatorname{Supp} D = p^{-1}(H_{[\ell]})$. Since $H_{[\ell]} \sim H_s$, we obtain $D \sim p^* H_s = A$.

By Definition 4.3, for every i, $p(x_i) \notin H_{[\ell]}$, so $x_i \notin \operatorname{Supp} D$. Since $x_i \in R_i$, it follows that $R_i \not\subset \operatorname{Supp} D$. Let T be a prime divisor component of D. Since $\dim Z \leq N-1 = \dim T$, we have $T \not\subset Z$: otherwise $T = R_i$ for some i, contradicting $R_i \not\subset \operatorname{Supp} D$. By the properties in Lemma 2.5, the restricted morphism

$$T \cap q^{-1}(O') \longrightarrow q(T) \cap O'$$

is a finite surjection and hence preserves dimension. Therefore,

$$\dim q(T) = \dim\big(q(T) \cap O'\big) = \dim\big(T \cap q^{-1}(O')\big) = \dim T = N-1.$$

□

Lemma 4.5. *Let D be the nonzero effective divisor chosen in Lemma 4.4. Then*

$$q^*(q_* D) - D \geq 0.$$

Proof. Write $D = \sum_{j=1}^{a} n_j D_j$, where $a \geq 1$, $n_j \in \mathbb{Z}_{>0}$, and the $D_j \subset Y$ are distinct prime divisors. By Lemma 4.4, each $q(D_j) \subset \mathbb{P}_t^N$ is also a prime Cartier divisor. By

$$q_* D = \sum_{j=1}^{a} n_j [\overline{\mathbb{Q}}(D_j) : \overline{\mathbb{Q}}(q(D_j))] q(D_j),$$

$q_* D$ is effective. Since q is dominant, $q^*(q(D_j))$ is effective, and $D_j \subset q^{-1}(q(D_j)) = \operatorname{Supp}(q^*(q(D_j)))$. Thus,

$$q^*(q_* D) = \sum_{j=1}^{a} n_j [\overline{\mathbb{Q}}(D_j) : \overline{\mathbb{Q}}(q(D_j))] q^*(q(D_j)) \geq \sum_{j=1}^{a} n_j D_j = D.$$

Since Y is normal, all coefficients of this Cartier divisor along prime divisors are nonnegative, and hence $q^*(q_* D) - D$ is an effective Cartier divisor. □

Proposition 4.6 (Degree and Linear Equivalence Class of the Constructed Divisor)**.** *With the notation of Lemma 4.5, let* $E = q^*(q_*D) - D \geq 0$ *and write* $\deg(q_*D) := (H_t^{N-1} \cdot q_*D)$*. Then*

$$1 \leq m = \deg(q_*D) \leq d^{N-1}, \qquad E \sim mB - A.$$

Proof. By the projection formula, $B = q^*H_t$, and $D \sim A$,

$$\deg(q_*D) = (H_t^{N-1} \cdot q_*D) = (B^{N-1} \cdot D) = (B^{N-1} \cdot A) = m.$$

Since q_*D is nonzero and effective, $m \geq 1$; Equation (4.1) gives $m \leq d^{N-1}$. A divisor of degree m on projective space is linearly equivalent to mH_t, so

$$E = q^*(q_*D) - D \sim mq^*H_t - A = mB - A.$$

□

Proof of Theorem 1.5. By Proposition 4.6, there exists an effective Cartier divisor $E \sim mB - A$, where $1 \leq m \leq d^{N-1}$. Therefore,

$$B - \frac{1}{m}A \sim_{\mathbb{Q}} \frac{1}{m}E$$

is pseudo-effective. By Proposition 1.3,

$$\mu_{\mathbb{P}^N}(\varphi) \geq \frac{1}{m} \geq d^{-(N-1)}.$$

□

When $N = 1$, $h(\varphi(P)) = dh(P) + O(1)$, $\mu_{\mathbb{P}^N}(\varphi) = d$. Combining the upper bound $\underline{\mu}_d(\mathbb{P}^N) \leq d^{-(N-1)}$ [17, Proposition 9], we obtain the following exact value of $\underline{\mu}_d(\mathbb{P}^N)$:

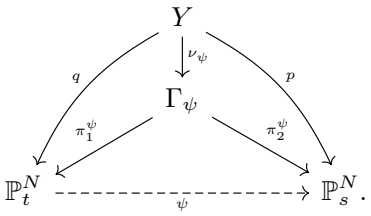
$$\underline{\mu}_d(\mathbb{P}^N) = \begin{cases} d, & N = 1, \\ d^{-(N-1)}, & N \geq 2. \end{cases}$$

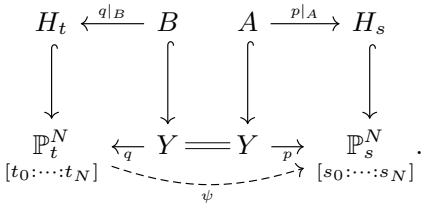
5 Equivalent Conditions for $\mu_{\mathbb{P}^N}(\varphi) = d^{-(N-1)}$

Proposition 5.1 (Birational case)**.** *If* φ *is birational, then*

$$\mu_{\mathbb{P}^N}(\varphi) = \frac{1}{m} = \frac{1}{\deg(\varphi^{-1})}. \tag{5.1}$$

Proof. Let $\psi = \varphi^{-1}$. Keeping the notation of Definition 4.1 and A, B, after interchanging $\mathbb{P}_s^N$ and $\mathbb{P}_t^N$, Y can still be regarded as the normalization of the graph of ψ. Then the commutative diagram says it all:





Since q is birational, the projection formula gives

$$m = (B^{N-1} \cdot A) = \left((q^*H_t)^{N-1} \cdot p^*H_s\right) = \left(H_t^{N-1} \cdot q_*p^*H_s\right) = \left(H_t^{N-1} \cdot \psi^*H_s\right) = \deg\psi.$$

By Theorem 1.5, we already have $\mu_{\mathbb{P}^N}(\varphi) \geq 1/m$. Let t be any rational number with $0 < t < \mu_{\mathbb{P}^N}(\varphi)$. By Corollary 2.7, $B - tA$ is pseudo-effective. Since B is nef and we have

$$0 \leq (B - tA) \cdot B^{N-1} = 1 - tm.$$

Hence $t \leq 1/m$. By the density of the rational numbers, it follows that $\mu_{\mathbb{P}^N}(\varphi) \leq 1/m$, which proves the assertion. □

Lemma 5.2. *If φ is not birational, then there exist an N-dimensional normal projective variety $\widetilde{Z}$ and surjective generically finite morphisms $r, s : \widetilde{Z} \to Y$, such that*

$$q \circ r = q \circ s, \qquad p \circ r \neq p \circ s.$$

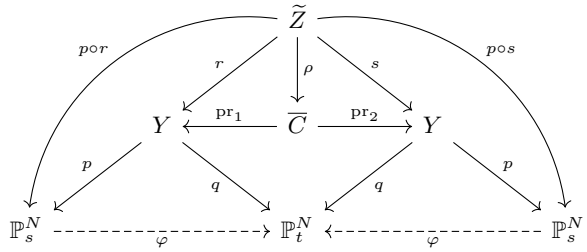
Proof. We continue to use the notation of Definition 4.1. $\deg q = [\overline{\mathbb{Q}}(Y) : \overline{\mathbb{Q}}(\mathbb{P}^N_t)] > 1$, q is projective, generically finite and separable. By [3, Exercise II.3.7] and [16, Proposition 5.51], after suitably shrinking the open set, we can choose a nonempty affine open set $V \subset \mathbb{P}^N_t$ such that $U = q^{-1}(V) \subset p^{-1}(\Omega)$, and $q|_U : U \to V$ is a finite étale morphism of degree $\deg q$. Let $\mathrm{pr}_1, \mathrm{pr}_2$ be the two coordinate projections, which are finite étale morphisms of degree $\deg q$.

$$\begin{array}{ccccc} U \times_V U & \xrightarrow{\mathrm{pr}_2} & U & \hookrightarrow & Y \\ \downarrow{\scriptstyle \mathrm{pr}_1} & & \downarrow{\scriptstyle q|_U} & & \downarrow{\scriptstyle q} \\ U & \xrightarrow[q|_U]{} & V & \hookrightarrow & \mathbb{P}^N_t. \end{array}$$

Let $\Delta_U \subset U \times_V U$ be the diagonal image. By [15, Proposition 2.15], Δ_U is a clopen connected component of $U \times_V U$, and $\mathrm{pr}_1|_{\Delta_U}$ is an isomorphism. Since $\deg(\mathrm{pr}_1) = \deg q > 1$, we may choose a connected component C different from Δ_U such that

$$C \subset (U \times_V U) \setminus \Delta_U. \tag{5.2}$$

Let $\overline{C} \subset Y \times_{\mathbb{P}^N_t} Y$ be the closure of C, and take its normalization $\rho : \widetilde{Z} \longrightarrow \overline{C}$. Write $r := \mathrm{pr}_1 \circ \rho$ and $s := \mathrm{pr}_2 \circ \rho$. They are surjective and generically finite, since r, s are proper and both projections $C \to U$ are finite étale surjections. Naturally, $q \circ r = q \circ s$. The corresponding commutative diagram is



By (5.2), choose a closed point $z \in \rho^{-1}(C)$ such that $r(z) \neq s(z) \in U \subset p^{-1}(\Omega)$. By Lemma 2.5 (3), p is an isomorphism on $p^{-1}(\Omega)$. Hence $p(r(z)) \neq p(s(z))$, so $p \circ r \neq p \circ s$. □

Lemma 5.3 (Pullbacks of a general hyperplane)**.** *There exists a nonempty open set $\mathcal{H}_2 \subset \mathcal{H}$, called* the second avoidance set*, such that for every $[\ell] \in \mathcal{H}_2$, $(p \circ r)^* H_{[\ell]}$ and $(p \circ s)^* H_{[\ell]}$ have no common prime components.*

Proof. Let $\Delta_{\mathbb{P}^N_s} \subset \mathbb{P}^N_s \times \mathbb{P}^N_s$ be the diagonal, and set

$$\Delta_{\widetilde{Z}} := (p \circ r, p \circ s)^{-1}(\Delta_{\mathbb{P}^N_s}).$$

It is a proper closed subset of $\widetilde{Z}$, so $\dim \Delta_{\widetilde{Z}} \le N-1$. Consider the following closed incidence subscheme:

$$\mathcal{I} := \big\{(z,[\ell]) \in \widetilde{Z} \times \mathcal{H} : p(r(z)) \in H_{[\ell]},\ p(s(z)) \in H_{[\ell]}\big\}.$$

Let $\pi_1 : \mathcal{I} \to \widetilde{Z}$ and $\pi_2 : \mathcal{I} \to \mathcal{H}$ be the projections. For each closed point $z \in \widetilde{Z}$ and each $[\ell] \in \mathcal{H}$, define the fibers:

$$\mathcal{I}_z := \pi_1^{-1}(z) \simeq \big\{[\ell] \in \mathcal{H} : \ell(p(r(z))) = \ell(p(s(z))) = 0\big\}.$$
$$\mathcal{I}_{[\ell]} := \pi_2^{-1}([\ell]) = \{z \in \widetilde{Z} : \ell(p(r(z))) = 0,\ \ell(p(s(z))) = 0\}.$$

We have the dichotomy: (1) If $z \notin \Delta_{\widetilde{Z}}$, then $p(r(z)) \neq p(s(z))$, so $\mathcal{I}_z \simeq \mathbb{P}^{N-2}$; (2) If $z \in \Delta_{\widetilde{Z}}$, then $p(r(z)) = p(s(z))$, so $\mathcal{I}_z \simeq \mathbb{P}^{N-1}$. Therefore,

$$\dim \mathcal{I} \le \max\big\{\dim(\widetilde{Z} \setminus \Delta_{\widetilde{Z}}) + N - 2,\ \dim \Delta_{\widetilde{Z}} + N - 1\big\} \le 2N-2.$$

By the fiber dimension inequality and upper semicontinuity [19, Lemma 37.30.5], define the nonempty open set

$$\mathcal{H}_2 := \big\{[\ell] \in \mathcal{H} : \dim \pi_2^{-1}([\ell]) \le N-2\big\}.$$

For every $[\ell] \in \mathcal{H}_2$, we have

$$\dim\big((p\circ r)^{-1}(H_{[\ell]}) \cap (p\circ s)^{-1}(H_{[\ell]})\big) = \dim \pi_2^{-1}([\ell]) \le N-2.$$

Thus, $(p\circ r)^* H_{[\ell]}$ and $(p\circ s)^* H_{[\ell]}$ have no common prime components. □

Corollary 5.4. *In Lemma 4.4, one can choose $\widetilde{D} = p^* H_{[\widetilde{\ell}]}$ so that $r^*\widetilde{D}$ and $s^*\widetilde{D}$ have no common prime components and $\widetilde{D} \sim A$.*

Proof. Take the nonempty open set $\mathcal{H}_2$ from Lemma 5.3. By Lemma 4.4, the avoidance set $\mathcal{H}_1$ in Definition 4.3 is also nonempty and open. Since $\mathcal{H}$ is irreducible, we may choose

$$[\widetilde{\ell}] \in (\mathcal{H}_1 \cap \mathcal{H}_2), \qquad \widetilde{D} := p^* H_{[\widetilde{\ell}]}.$$

By Lemma 5.3,

$$r^*\widetilde{D} = (p\circ r)^* H_{[\widetilde{\ell}]}, \qquad s^*\widetilde{D} = (p\circ s)^* H_{[\widetilde{\ell}]}$$

have no common prime components. Clearly, $\widetilde{D} \sim A$. □

Lemma 5.5 (Non-birational case)**.** *If φ is not birational, then*

$$mB - A \text{ is big}, \qquad \mu(\varphi) > \frac{1}{m}.$$

Proof. Choose $\widetilde{D} = p^* H_{[\widetilde{\ell}]}$ as in Corollary 5.4, so that $r^*\widetilde{D}$ and $s^*\widetilde{D}$ have no common prime components, and set $E := q^*(q_*\widetilde{D}) - \widetilde{D}$. By Lemma 4.5 and Proposition 4.6, $E \ge 0$ and $E \sim mB - A$.

Since $q\circ r = q \circ s$, we have

$$r^*(\widetilde{D}+E) = (q\circ r)^*(q_*\widetilde{D}) = (q\circ s)^*(q_*\widetilde{D}) = s^*(\widetilde{D}+E).$$

Since r, s are dominant, $r^*E, s^*E \ge 0$. Write $s^*\widetilde{D} = \sum_{i=1}^{b} b_i T_i$, where $b \ge 1$, $b_i \in \mathbb{Z}_{>0}$, and the $T_i \subset \widetilde{Z}$ are distinct prime divisors. The coefficient of $r^*\widetilde{D}$ along each T_i is zero. Therefore,

$$\operatorname{coeff}_{T_i}(r^*E) = b_i + \operatorname{coeff}_{T_i}(s^*E) \ge b_i.$$

Together with $r^*E \geq 0$, this gives

$$r^*E \geq \sum_{i=1}^{b} b_i T_i = s^*\widetilde{D} = (p \circ s)^* H_{[\widetilde{\ell}]}.$$

By the volume formula for generically finite morphisms [5, Lemma 4.3], we have

$$\deg(r)\,\mathrm{vol}_Y(E) = \mathrm{vol}_{\widetilde{Z}}(r^*E) \geq \mathrm{vol}_{\widetilde{Z}}\big((p \circ s)^* H_{[\widetilde{\ell}]}\big) = \deg(p \circ s) > 0.$$

Hence $E \sim mB - A$ and $E \in \mathrm{Big}(Y)$.

By the openness of the big cone [7, Corollary 2.2.24], there exists $\varepsilon > 0$ such that $E - \varepsilon A \in \mathrm{Big}(Y)$ and

$$B - \frac{1+\varepsilon}{m} A \sim_{\mathbb{R}} \frac{1}{m}(E - \varepsilon A) \in \mathrm{Big}(Y) \subset \mathrm{PsEff}(Y).$$

Then Proposition 1.3 gives

$$\mu_{\mathbb{P}^N}(\varphi) \geq \frac{1+\varepsilon}{m} > \frac{1}{m}.$$

□

Lemma 5.6 (Equality conditions for the degree bound)**.** *Retain the notation of Definition 4.1 and suppose that $N \geq 2$. Then*

$$m = d^{N-1} \quad \text{if and only if} \quad \dim \mathrm{Ind}(\varphi) \leq 0.$$

Proof. Let $\mathcal{J}$ be the ideal sheaf on $\mathbb{P}^N_s$ generated by $F_0, \ldots, F_N$ with vanishing locus $\mathrm{Ind}(\varphi)$. The graph closure Γ is naturally isomorphic to $\mathrm{Bl}_{\mathcal{J}}(\mathbb{P}^N_s)$, with blow-up morphism π_1. Let E_Γ be the exceptional divisor of π_1. By [19, Tag 02OS] and [2, §4.4], it is an effective Cartier divisor and satisfies

$$\mathcal{J}\mathcal{O}_\Gamma = \mathcal{O}_\Gamma(-E_\Gamma), \qquad \mathrm{Supp}\, E_\Gamma = \pi_1^{-1}\big(\mathrm{Ind}(\varphi)\big), \qquad \pi_2^* H_t \sim d\pi_1^* H_s - E_\Gamma.$$

Let $\widetilde{\mathcal{J}}$ be the ideal sheaf of E_Γ in Γ, and define $E_0 := \nu^* E_\Gamma$. We obtain

$$\widetilde{\mathcal{J}}\mathcal{O}_Y = \mathcal{O}_Y(-E_0), \qquad \mathrm{Supp}\, E_0 = p^{-1}\big(\mathrm{Ind}(\varphi)\big), \qquad B \sim dA - E_0.$$

Let $E_1, \ldots, E_{a'}$ be all the prime components of E_0, and let Σ_0 be the set of points consisting of all zero-dimensional irreducible components of $\mathrm{Ind}(\varphi)$. By Noetherianity, both are finite. Choose closed points $y_i \in E_i$. Then $\Sigma_0 \cup \{p(y_1), \ldots, p(y_{a'})\}$ is also finite, so we may choose a hyperplane $H' \subset \mathbb{P}^N_s$ such that

$$H' \cap \big(\Sigma_0 \cup \{p(y_1), \ldots, p(y_{a'})\}\big) = \varnothing.$$

Thus $E_i \not\subset p^{-1}(H')$ for all i, so $T_0 := E_0 \cdot p^* H' \sim E_0 \cdot A$ is an effective $(N-2)$-dimensional cycle, and

$$\mathrm{Supp}\, T_0 = \mathrm{Supp}\, E_0 \cap p^{-1}(H') = p^{-1}\big(H' \cap \mathrm{Ind}(\varphi)\big).$$

We have the dichotomy: (1) If $\dim \mathrm{Ind}(\varphi) \leq 0$, then $\mathrm{Supp}\, T_0 = \varnothing$, and hence $T_0 = 0$; (2) If $\dim \mathrm{Ind}(\varphi) \geq 1$, then $\mathrm{Supp}\, T_0 \neq \varnothing$, and hence $T_0 \neq 0$. Therefore

$$T_0 = 0 \quad \Longleftrightarrow \quad \dim \mathrm{Ind}(\varphi) \leq 0.$$

By $E_0 \sim dA - B$ and $p^*H' \sim A$, we have

$$\begin{aligned} d^{N-1} - m &= \big(A \cdot ((dA)^{N-1} - B^{N-1})\big) \\ &= \sum_{j=0}^{N-2} d^{N-2-j}\big(A \cdot (dA - B) \cdot A^{N-2-j} \cdot B^j\big) \\ &= \sum_{j=0}^{N-2} d^{N-2-j}\big(T_0 \cdot A^{N-2-j} \cdot B^j\big). \end{aligned} \tag{$*$}$$

Since T_0 is effective and A, B are nef, $\big(T_0 \cdot A^{N-2-j} \cdot B^j\big) \geq 0$ for all j. On the other hand,

$$\big(T_0 \cdot (A+B)^{N-2}\big) = \sum_{j=0}^{N-2} \binom{N-2}{j} \big(T_0 \cdot A^{N-2-j} \cdot B^j\big) \geq 0. \tag{$**$}$$

By nonnegativity, ($*$) and ($**$) vanish simultaneously. Moreover, since $A + B$ is ample, $T_0 \cdot (A+B)^{N-2}$ is strictly positive for $T_0 \neq 0$ [7, Theorem 1.2.23]. Thus

$$m = d^{N-1} \iff \big(T_0 \cdot (A+B)^{N-2}\big) = 0 \iff T_0 = 0 \iff \dim \mathrm{Ind}(\varphi) \leq 0.$$

□

Proof of Theorem 1.6. By the proof of Theorem 1.5, $\mu_{\mathbb{P}^N}(\varphi) \geq 1/m \geq d^{-(N-1)}$. Therefore, (1) $\iff \mu_{\mathbb{P}^N}(\varphi) = 1/m$ and $m = d^{N-1}$. By Proposition 5.1 and Lemma 5.5, (1) $\iff$ (2). By Lemma 5.6, (2) $\iff$ (3).

When $N = 2$, $\gcd(F_0, F_1, F_2) = 1$ ensures that the indeterminacy locus has dimension at most zero, proving the last assertion. □

Remark 5.7. When $\varphi\colon \mathbb{P}^2 \dashrightarrow \mathbb{P}^2$ is a birational map of degree $d \geq 2$, Theorem 1.6 gives

$$\mu_{\mathbb{P}^2}(\varphi) = \frac{1}{d} < 1 \leq \limsup_{n\to\infty} \max\{1, h(\varphi^n(P))\}^{1/n}$$

for every point P whose forward orbit is well-defined. Thus the height expansion coefficient is not the same as the height growth rate.

The possible values of the height expansion coefficient are also of interest. Silverman [17, Question 11] asks whether $\mu_{\mathbb{P}^N}(\varphi)$ is always a rational power of $\deg \varphi$. In fact, Lee's example [8, Example 7.7] already yields a negative answer to this question. We show that every positive rational number can be realized as the height expansion coefficient of a dominant rational self-map of $\mathbb{P}^2$.

Proposition 5.8 (Realization of arbitrary positive rational numbers)**.** *For any positive integers a, b, define the rational map*

$$\varphi_{a,b} : \mathbb{P}^2 \dashrightarrow \mathbb{P}^2, \qquad [X : Y : Z] \longmapsto \big[X^a Z^{a(b-1)} : X^{a(b-1)} Y^a : Z^{ab}\big].$$

Then $\varphi_{a,b}$ is dominant, and

$$\deg(\varphi_{a,b}) = ab, \qquad \mu_{\mathbb{P}^2}(\varphi_{a,b}) = \frac{a}{b}.$$

Consequently, every positive rational number can be realized as the height expansion coefficient of a dominant rational self-map of $\mathbb{P}^2_{\overline{\mathbb{Q}}}$.

Proof. Let

$$\beta_b : \mathbb{P}^2 \dashrightarrow \mathbb{P}^2, \qquad [X : Y : Z] \longmapsto [XZ^{b-1} : X^{b-1}Y : Z^b].$$

Clearly β_b is birational, and its inverse has the homogeneous representation $\beta_b^{-1}[X : Y : Z] = [X^b : YZ^{b-1} : X^{b-1}Z]$, so $\deg(\beta_b^{-1}) = b$. By Proposition 5.1,

$$\mu_{\mathbb{P}^2}(\beta_b) = \frac{1}{\deg(\beta_b^{-1})} = \frac{1}{b}.$$

Now let

$$S_a : \mathbb{P}^2 \longrightarrow \mathbb{P}^2, \qquad [X : Y : Z] \longmapsto [X^a : Y^a : Z^a].$$

By construction, $\varphi_{a,b} = S_a \circ \beta_b$. Since S_a is finite and surjective and β_b is birational, $\varphi_{a,b}$ is dominant. It is easy to see that

$$\deg(\varphi_{a,b}) = ab, \qquad h(\varphi_{a,b}(P)) = h(S_a(\beta_b(P))) = a\, h(\beta_b(P)).$$

Together with Definition 1.2, this gives

$$\mu_{\mathbb{P}^2}(\varphi_{a,b}) = a\, \mu_{\mathbb{P}^2}(\beta_b) = \frac{a}{b}.$$

□

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